

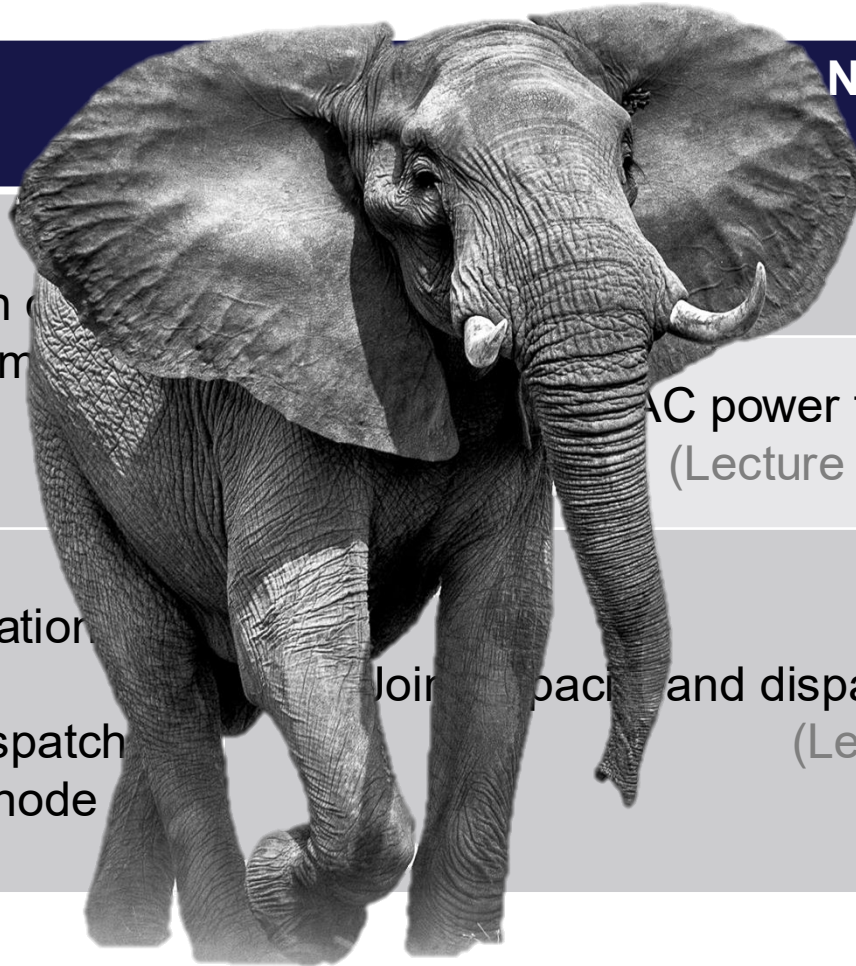
46770 Integrated Energy Grids

Aleksander Grochowicz

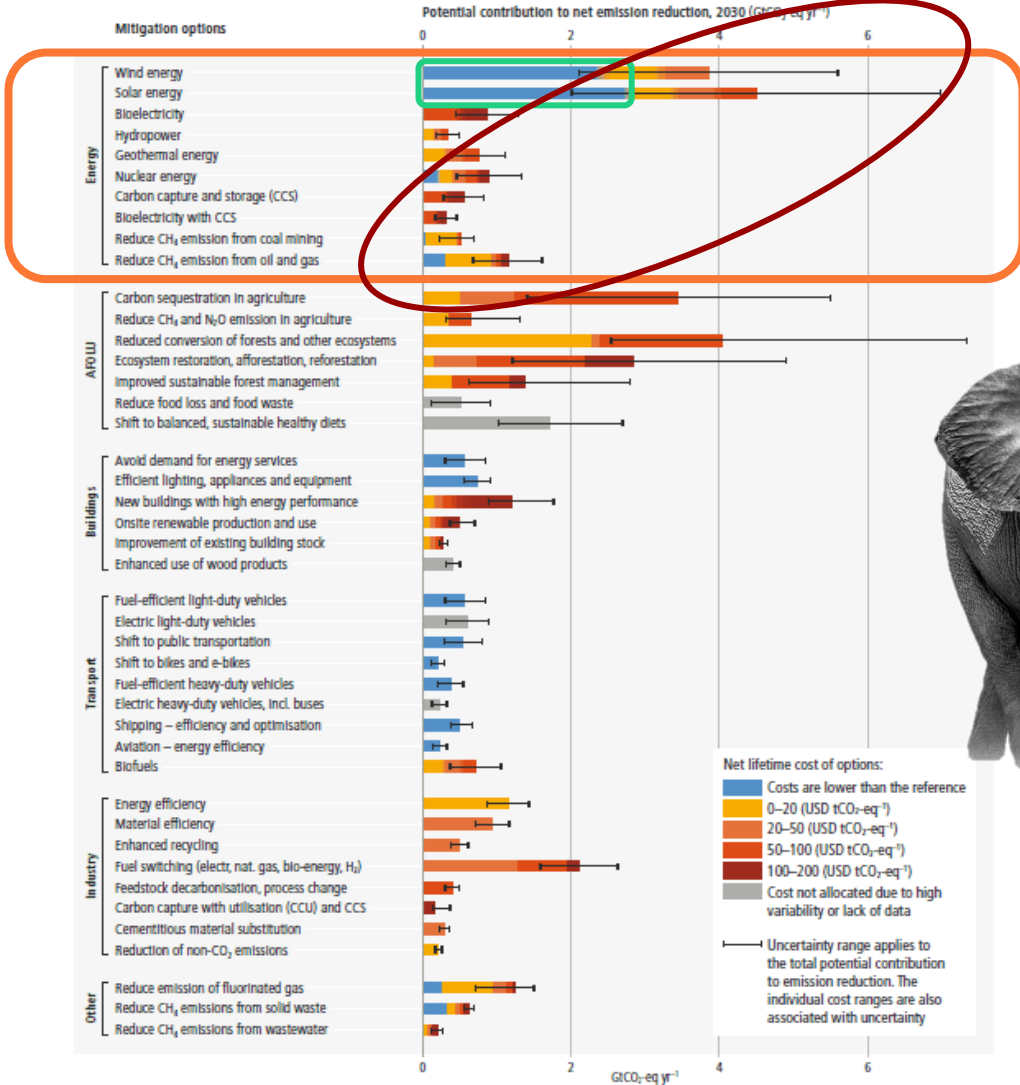
# Lecture 9 – Limiting CO<sub>2</sub> emissions

# Types of optimisation problems and course structure

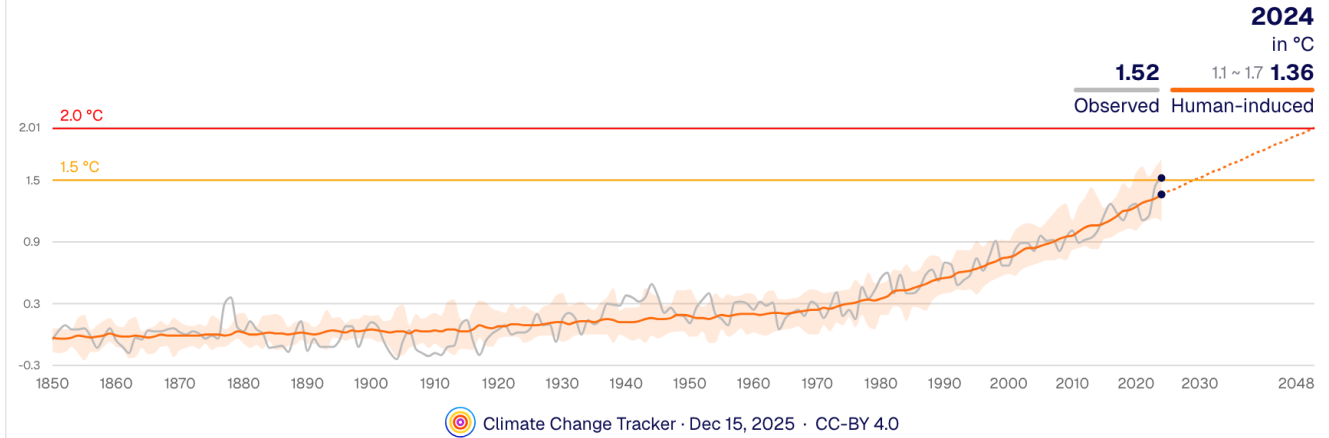
|                     | One node                                                                                      | Network                                                            |                      |                       |
|---------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------|----------------------|-----------------------|
| One time step       | Economic dispatch of one node<br>One-node dispatch optimisation (Lecture 2)                   | AC power flow (Lecture 6)                                          | Gas flow (Lecture 7) | Heat flow (Lecture 8) |
| Multiple time steps | Multi-period optimisation<br>Joint capacity and dispatch optimisation in one node (Lecture 3) | Joint capacity and dispatch optimisation in a network (Lecture 10) |                      |                       |



# The elephant in the room

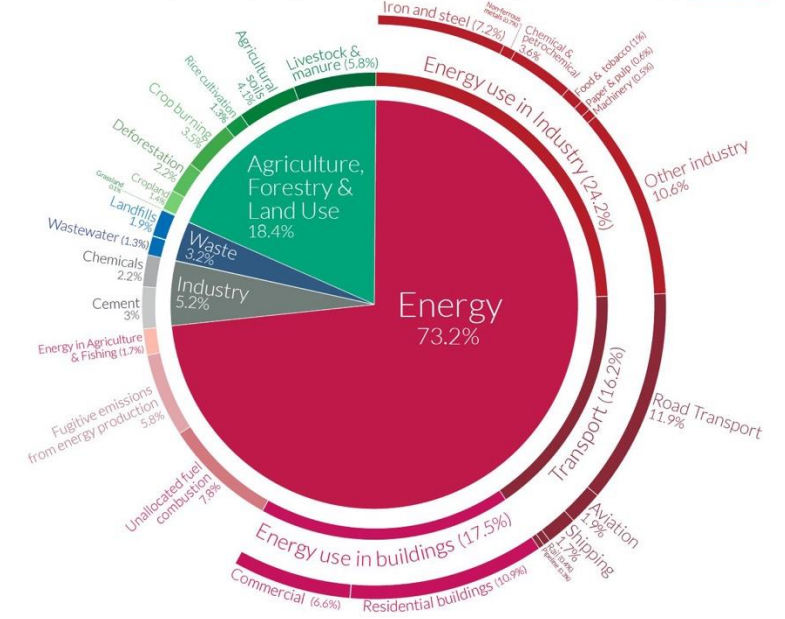


## Historic Human-Induced and Observed Global Warming



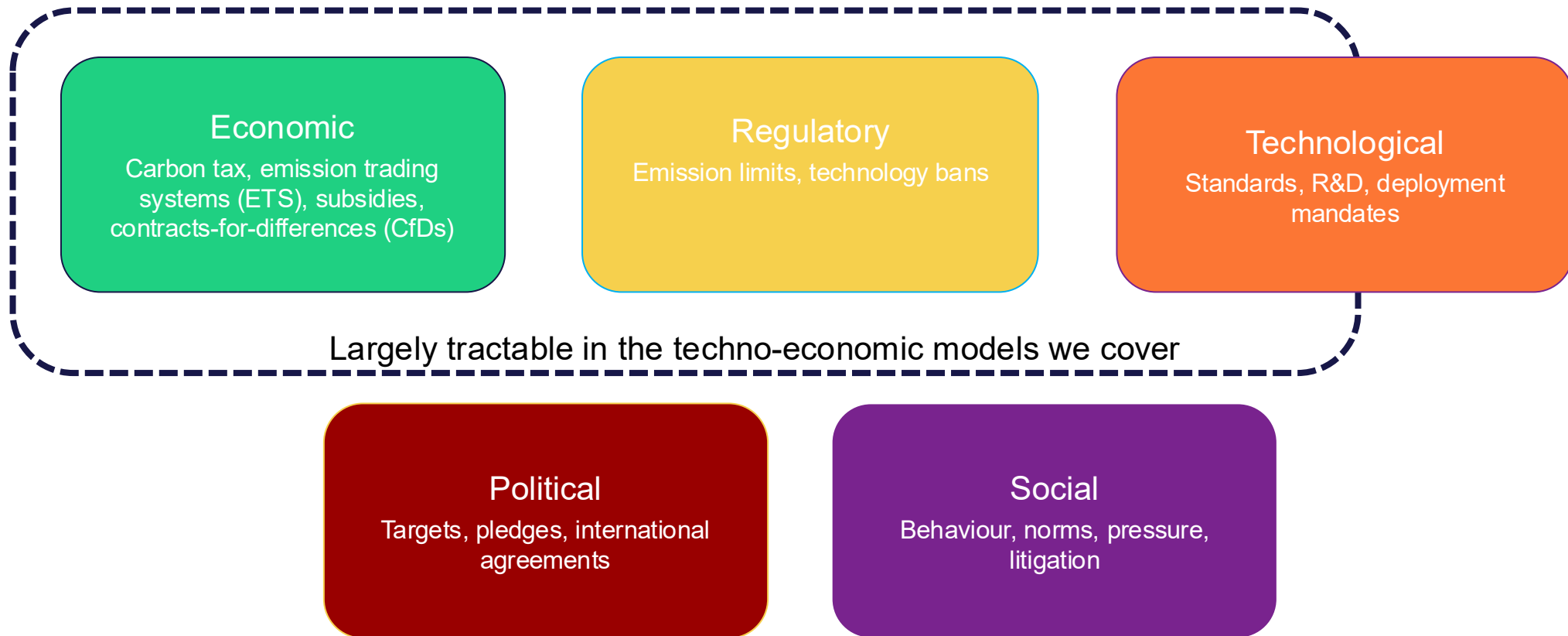
## Global greenhouse gas emissions by sector

This is shown for the year 2016 – global greenhouse gas emissions were 49.4 billion tonnes CO<sub>2</sub>eq.



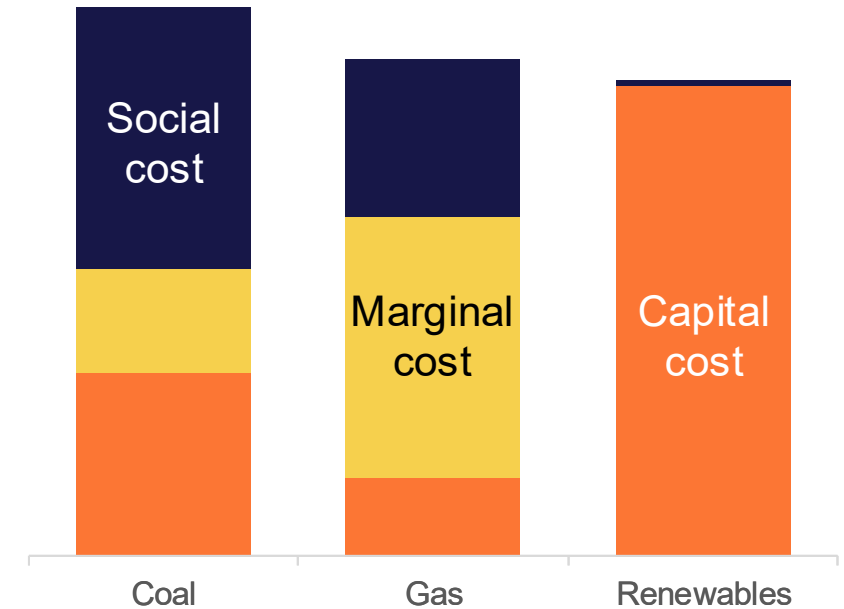
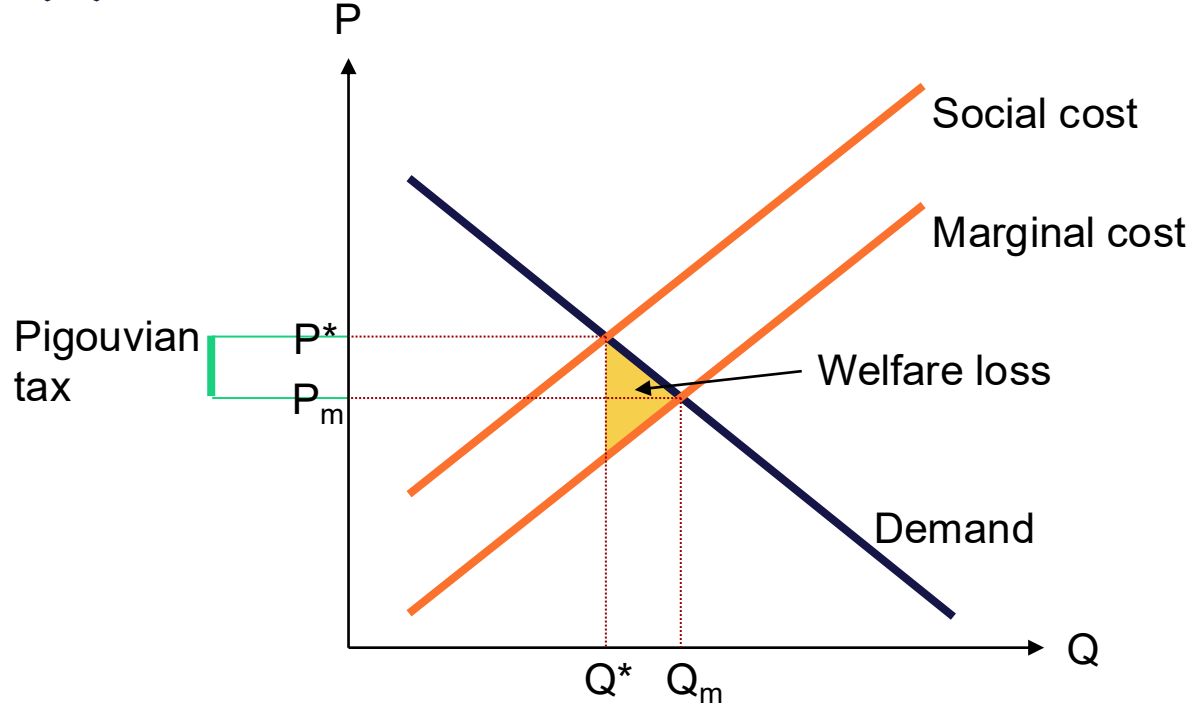
OurWorldinData.org – Research and data to make progress against the world's largest problems.  
Source: Climate Watch, the World Resources Institute (2020). Licensed under CC-BY by the author Hannah Ritchie (2020).

# Different policies exist to accelerate the transition



\* By no means exhaustive, domains interact.

# Why do markets fail?



An **externality** is a cost or benefit imposed on a third party that is not reflected in the market prices. Examples: clean air, neighbour's loud music, CO<sub>2</sub> emissions, public goods.

The Pigouvian tax internalises the costs/benefits of the externality to society. How to set it?

# Learning goals for this lecture

- Demonstrate cost recovery in electricity markets.
- Explain how the CO<sub>2</sub> price affects the cost recovery in electricity markets.
- Interpret the meaning of the Lagrange/KKT multiplier associated with the CO<sub>2</sub> emissions constraints and analyse its value.
- Formulate a joint capacity and dispatch optimisation with CO<sub>2</sub> emissions constraints on a computer.
- Define financial discount rate and social discount rate.
- Calculate the cost annualisation factor.
- Describe the different approach that classic economy has taken to include climate change in economic optimisation.

# Cost recovery in electricity markets

# Cost recovery in optimised markets (“non-profit rule”)

Recall from Lecture 3 that our optimisation variables are the energy  $g_{s,t}$  generated by every generator  $s$  in every time step  $t$  and the installed capacity  $G_s$  of every generator

$$\left\{ \begin{array}{l} \min_{g_{s,t}, G_s} \left[ \sum_s c_s G_s + \sum_{s,t} o_s g_{s,t} \right] \\ \text{subject to:} \\ \sum_s g_{s,t} - d_t = 0 \leftrightarrow \lambda_t \\ -g_{s,t} + G_s \geq 0 \leftrightarrow \overline{\mu_{s,t}} \end{array} \right.$$

$$0 = \frac{\partial \mathcal{L}}{\partial g_{s,t}} = \frac{\partial f}{\partial g_{s,t}} - \sum_i \lambda_i \frac{\partial h_i}{\partial g_{s,t}} - \sum_j \mu_j \frac{\partial g_j}{\partial g_{s,t}} = o_s - \lambda_t^* + \overline{\mu_{s,t}^*} = 0 \quad \Rightarrow o_s = \lambda_t^* - \overline{\mu_{s,t}^*}$$

$$0 = \frac{\partial \mathcal{L}}{\partial G_s} = \frac{\partial f}{\partial G_s} - \sum_i \lambda_i \frac{\partial h_i}{\partial G_s} - \sum_j \mu_j \frac{\partial g_j}{\partial G_s} = c_s - \sum_t \overline{\mu_{s,t}^*} \cdot (1) = 0 \quad \Rightarrow c_s = \sum_t \overline{\mu_{s,t}^*}$$

# Cost recovery in optimised markets (“non-profit rule”)

Total cost for generator  $s$

$$c_s G_s^* + \sum_t o_s g_{s,t}^* = c_s G_s^* + \sum_t (\lambda_t^* - \overline{\mu_{s,t}^*}) g_{s,t}^*$$

$$c_s G_s^* + \sum_t o_s g_{s,t}^* = c_s G_s^* + \sum_t \lambda_t^* g_{s,t}^* - \underbrace{\sum_t \overline{\mu_{s,t}^*}}_{c_s} \underbrace{g_{s,t}^*}_{G_s^*}$$

$$c_s G_s^* + \sum_t o_s g_{s,t}^* = \underbrace{c_s G_s^*}_0 + \sum_t \lambda_t^* g_{s,t}^* - \underbrace{c_s G_s^*}_0$$

$$c_s G_s^* + \sum_t o_s g_{s,t}^* = \sum_t \lambda_t^* g_{s,t}^*$$

This is only true if the constraint is binding, but otherwise  $\overline{\mu_{s,t}^*} = 0$ .

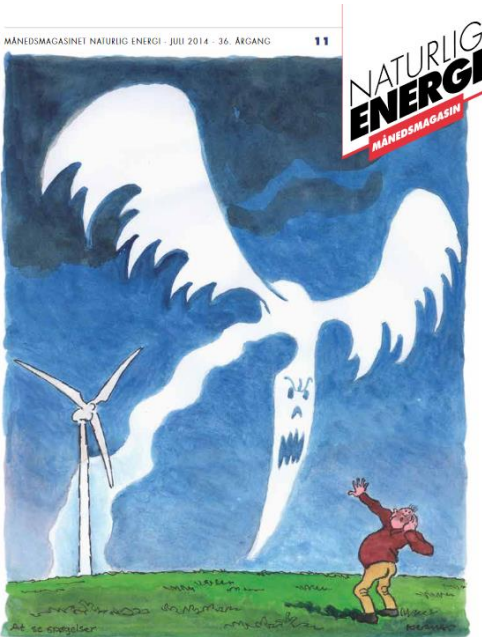
The generator costs (fixed and variable) are **exactly fully-recovered** from market revenues (generation times market price). This is known as **long-term market equilibrium** or **non-profit rule**.

If a price cap is set in the market, this is not true anymore. Some countries set a price cap and use capacity markets to compensate for the unsuccessful recovery of capital cost.

# Assumptions included in the model / Limitations

The optimisation problem makes three assumptions:

- Perfect competition: no generator has the power to impose artificially high electricity prices.
- Long-term market equilibrium: thanks to market revenues every technology recovers exactly its cost (fixed & variable) throughout its lifetime. This requires that, for some hours, electricity prices are higher than the marginal cost of the most expensive technology.
- Perfect foresight through the year: This might not be true (e.g., the model assumes that reservoir hydropower can be used at the optimal time).



We can include additional constraints to represent social acceptance issues (e.g., limit transmission expansion or the maximum capacity for onshore wind).

$$\sum_l \text{Capacity}_l \cdot \text{Length}_l < \text{Transmission cap}$$

We can search for alternative solutions with higher public acceptance and quantify the economic impacts (→ Lecture 13).

# Global CO<sub>2</sub> constraint

# Global CO<sub>2</sub> constraint

We can add a constraint to limit the total CO<sub>2</sub> emissions in the system

$$\sum_{s,t} \frac{\varepsilon_s}{\eta_s} g_{s,t} \leq \text{CAP}_{\text{CO}_2} \leftrightarrow \mu_{\text{CO}_2}$$

where  $\varepsilon_s$  are the specific emissions per technology (in tCO<sub>2</sub>/MWh<sub>thermal</sub>), and  $\eta_s$  is the efficiency of the generator (in MWh<sub>electricity</sub> / MWh<sub>thermal</sub>).

$\mu_{\text{CO}_2}$  is the Lagrange or Karush-Kuhn-Tucker (KKT) multiplier associated with the CO<sub>2</sub> constraint. It represents the change in the objective function at the optimal solution, with respect to a small change in the constraint.

Small change in constraint: CO<sub>2</sub> limit\* = CO<sub>2</sub> limit + 1 tonne

Change in objective function: System cost\* = System cost +  $\Delta$  System cost

Here  $\mu_{\text{CO}_2}$  represents the cost of 1 tonne of CO<sub>2</sub>.

- If  $\mu_{\text{CO}_2} = 0$ , the constraint is not binding.
- If  $\mu_{\text{CO}_2} \neq 0$ , we would obtain the same solution if we removed the constraint and added a CO<sub>2</sub> price equal to  $\mu_{\text{CO}_2}$ .

$\mu_{\text{CO}_2}$  is also known as **CO<sub>2</sub> price, CO<sub>2</sub> tax, shadow price or marginal abatement cost**.

# Setting a global CO<sub>2</sub> constraint is equivalent to adding a CO<sub>2</sub> tax

If we repeat the demonstration of cost recovery in optimised markets (“non-profit rule”), including the CO<sub>2</sub> constraint:

$$\left\{ \begin{array}{l} \min_{g_s, G_s} \left[ \sum_s c_s G_s + \sum_{s,t} o_s g_{s,t} \right] \\ \text{subject to:} \\ \sum_s g_{s,t} - d_t = 0 \leftrightarrow \lambda_t \\ -g_{s,t} + G_s \geq 0 \leftrightarrow \overline{\mu_{s,t}} \\ -\sum_{s,t} \frac{\varepsilon_s}{\eta_s} g_{s,t} + \text{CAP}_{\text{CO}_2} \geq 0 \leftrightarrow \mu_{\text{CO}_2} \end{array} \right.$$

The derivative of the Lagrangian with respect to the generation gets an extra factor:

$$0 = \frac{\partial \mathcal{L}}{\partial g_{s,t}} = \frac{\partial f}{\partial g_{s,t}} - \sum_i \lambda_i \frac{\partial h_i}{\partial g_{s,t}} - \sum_j \mu_j \frac{\partial g_j}{\partial g_{s,t}} = o_s - \lambda_t^* + \overline{\mu_{s,t}^*} + \mu_{\text{CO}_2} \frac{\varepsilon_s}{\eta_s} = 0 \quad \Rightarrow o_s = \lambda_t^* - \overline{\mu_{s,t}^*} - \mu_{\text{CO}_2} \frac{\varepsilon_s}{\eta_s}$$

# Setting a global CO<sub>2</sub> constraint is equivalent to adding a CO<sub>2</sub> tax

We repeat the demonstration of cost recovery in optimised markets (“non-profit rule”):

$$c_s G_s^* + \sum_t o_s g_{s,t}^* = c_s G_s^* + \sum_t (\lambda_t^* - \overline{\mu_{s,t}^*} - \mu_{\text{CO}_2} \frac{\varepsilon_s}{\eta_s}) g_{s,t}^*$$

$$c_s G_s^* + \sum_t o_s g_{s,t}^* = c_s G_s^* + \sum_t (\lambda_t^* - \mu_{\text{CO}_2} \frac{\varepsilon_s}{\eta_s}) g_{s,t}^* - \underbrace{\sum_t \overline{\mu_{s,t}^*} - g_{s,t}^*}_{c_s G_{s,t}^*}$$

This is only true if the constraint is binding, but otherwise  $\overline{\mu_{s,t}^*} = 0$ .

$$c_s G_s^* + \sum_t o_s g_{s,t}^* = \underbrace{c_s G_s^*}_0 + \sum_t (\lambda_t^* - \mu_{\text{CO}_2} \frac{\varepsilon_s}{\eta_s}) g_{s,t}^* - \underbrace{c_s G_s^*}_0$$

$$c_s G_s^* + \sum_t o_s g_{s,t}^* = \sum_t \underbrace{(\lambda_t^* - \mu_{\text{CO}_2} \frac{\varepsilon_s}{\eta_s}) g_{s,t}^*}_{\text{Market revenues minus cost of CO}_2 \text{ price times generation}}$$

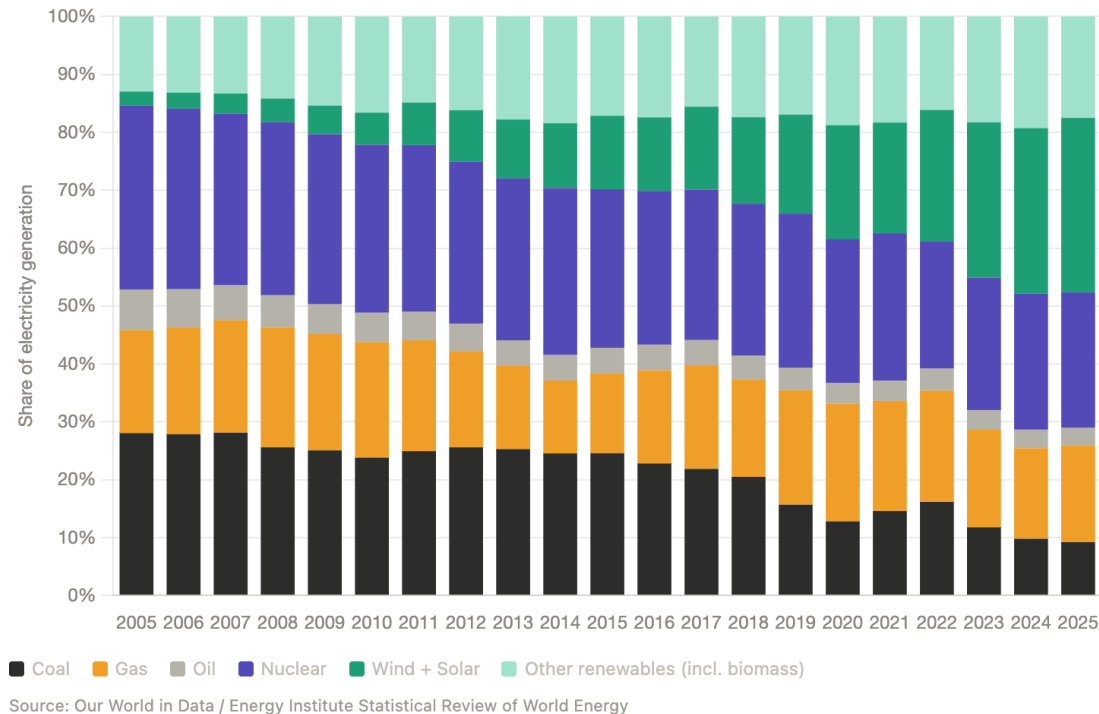
Market revenues minus cost of CO<sub>2</sub> price times generation

This shows that **setting a global CO<sub>2</sub> constraint is equivalent to adding a CO<sub>2</sub> price**, i.e. substituting  $o_s$  by  $o_s + \mu_{\text{CO}_2} \frac{\varepsilon_s}{\eta_s}$ .

# CO<sub>2</sub> pricing in the EU

- Emission trading system (EU ETS 1) introduced in 2005
- 50% emission reduction in involved sectors between 2000 and 2050
- For electricity generation: point sources of carbon emissions, measurable, immobile
- Currently ETS includes power and heat generation, energy-intensive industry sectors, commercial aviation.

Extending the same logic to industry and other sectors is more complex. Issues include existence of alternatives, carbon leakage, politics.



Emissions cap  
(decreasing)

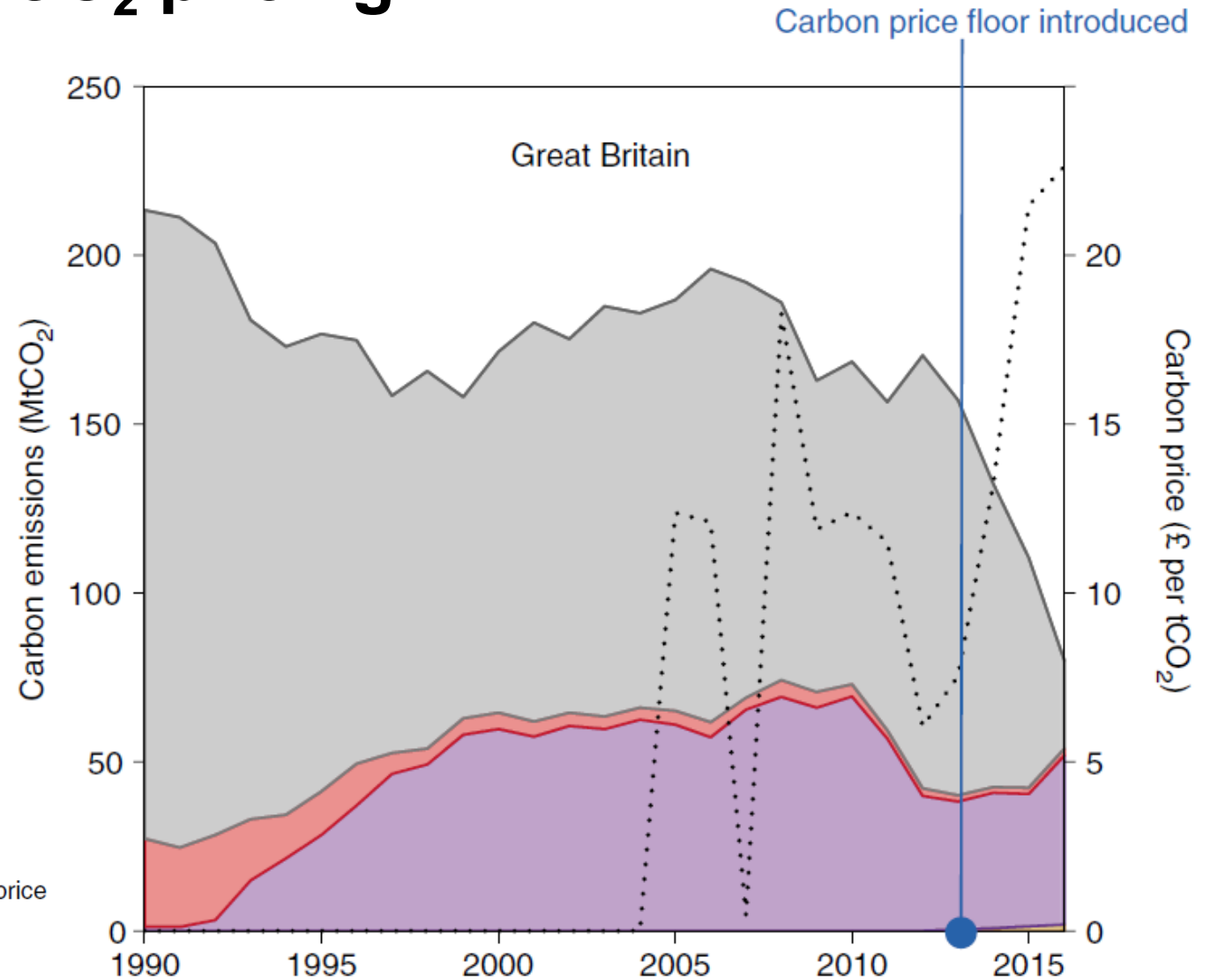
Issue of certificates  
(either sold or  
allocated freely)

Traded freely

Market price

# An example of effective CO<sub>2</sub> pricing

“Great Britain carbon emissions fell drastically and quickly not due to a surge in low-carbon nuclear or renewable sources. Instead, it was the impact of fuel switching from coal to natural gas generation incentivized through a stable and strong carbon price.”



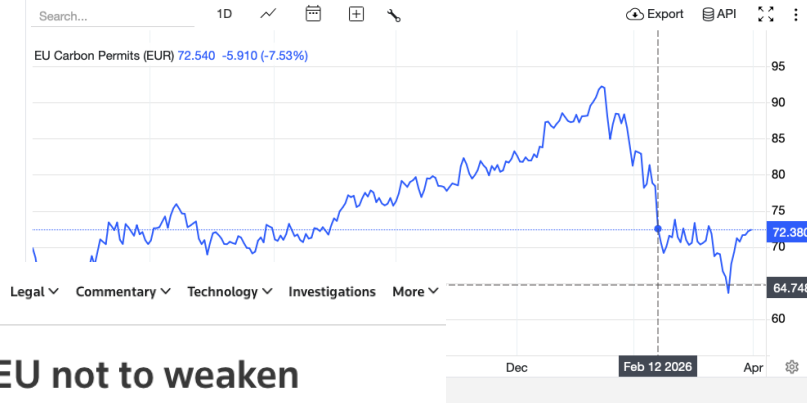
Source: Wilson and Staffell, [Rapid fuel switching from coal to natural gas through effective carbon pricing](#), Nature (2018)

# Credibility is key

- A CO<sub>2</sub> constraint (or CO<sub>2</sub> tax) adds a shadow price to every unit of emissions pricing the externality.
- Price signals only work if actors believe it will hold.
- Price drives investment if plans are long-sighted.
- Carbon prices will be paid today, and cheap decarbonisation and low-hanging fruits will be prioritised.
- Collapse of credibility: actors revert to short-run thinking (delayed investment, stranded assets).

Read more: Sitarz et al, Nature, 2024. [EU carbon prices signal high policy credibility and farsighted actors](#)

## EU Carbon Prices Plunge as Merz Signals Openness to Soften Rules



## Eight countries warn EU not to weaken carbon market, document shows

By Kate Abnett

March 12, 2026 3:54 PM GMT+1 · Updated March 12, 2026

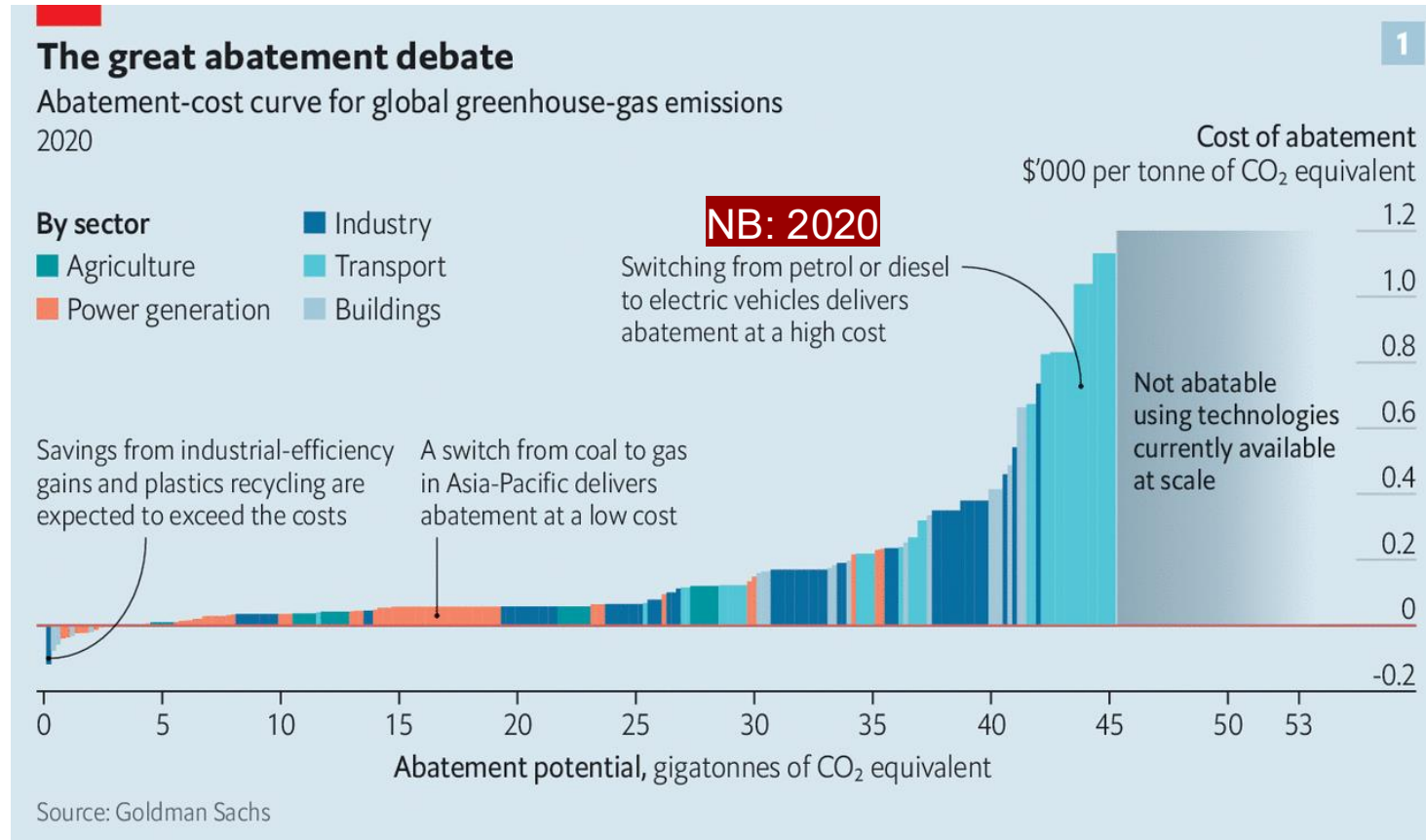


## Italy calls for suspension of carbon price in major attack on EU climate policy

The call would allow businesses to pollute for free, undermining the bloc's longstanding emissions trading scheme.

# CO<sub>2</sub> pricing and abatement costs

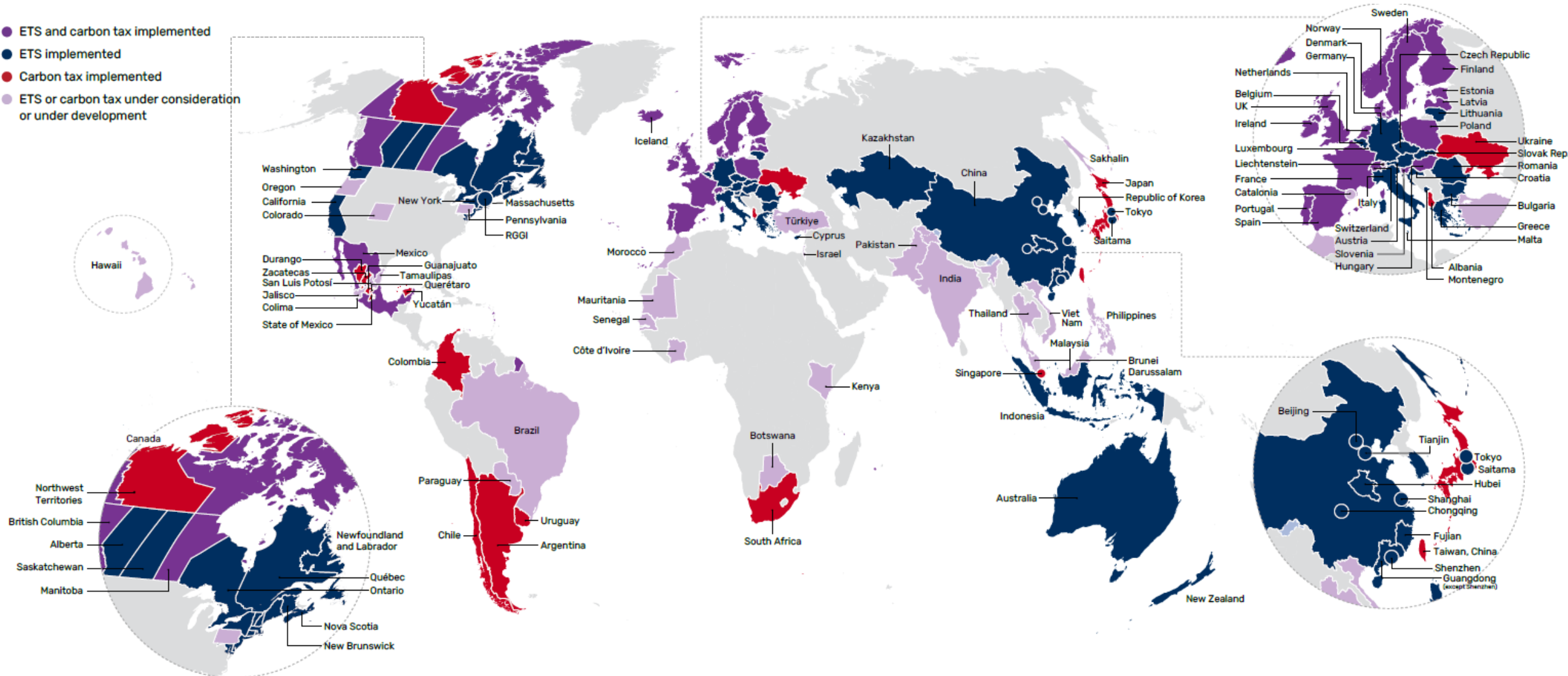
Extending the same logic to industry and other sectors is more complex. Issues include existence of alternatives, carbon leakage, politics.



The Economist

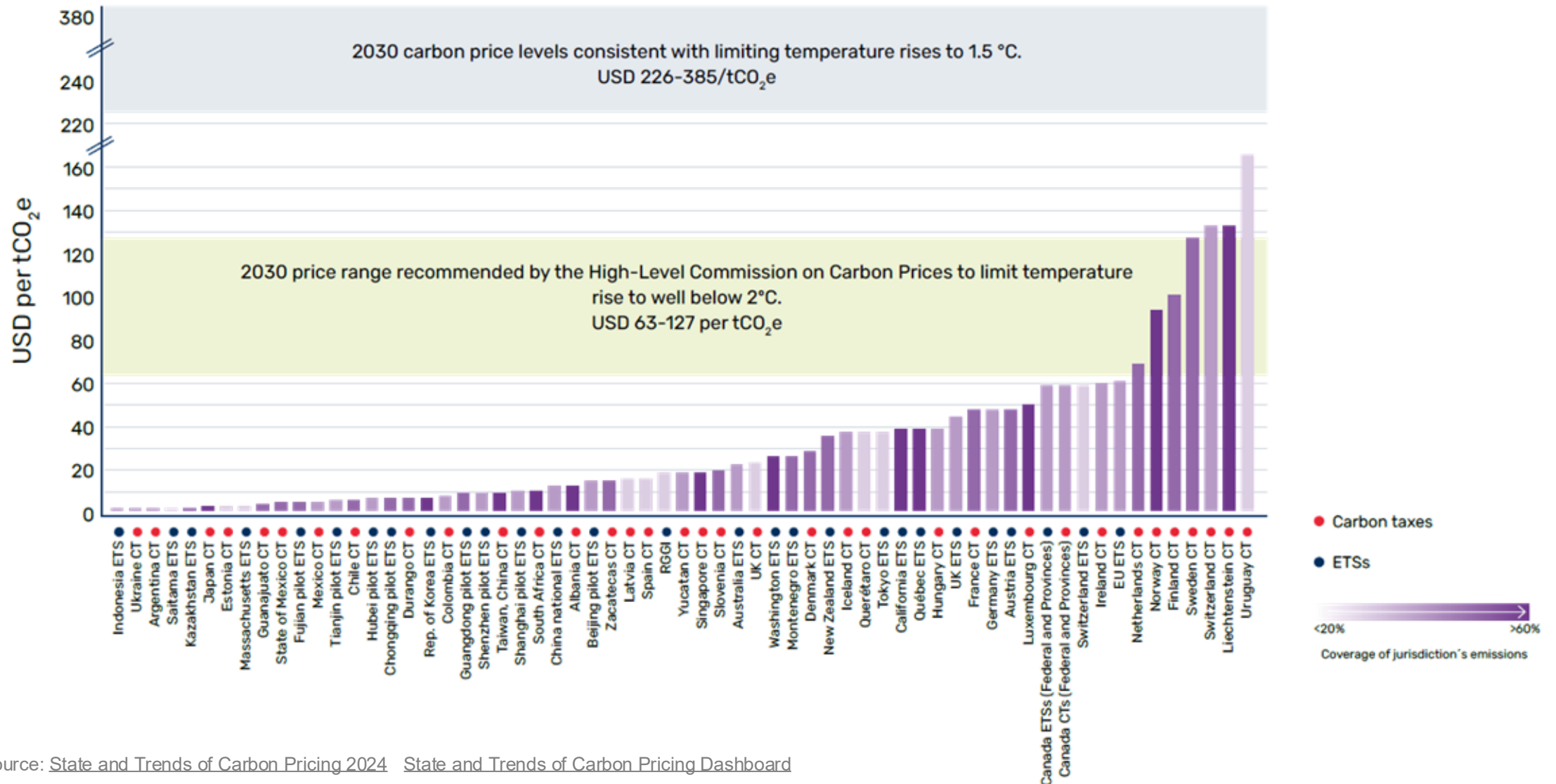
# Existing CO<sub>2</sub> tax in different countries

- ETS and carbon tax implemented
- ETS implemented
- Carbon tax implemented
- ETS or carbon tax under consideration or under development



Source: [State and Trends of Carbon Pricing 2024](#) [State and Trends of Carbon Pricing Dashboard](#)

# CO<sub>2</sub> price levels continue to fall short of the ambition needed to achieve the Paris Agreement goals



Source: [State and Trends of Carbon Pricing 2024](#) [State and Trends of Carbon Pricing Dashboard](#)

# Cost annualisation and financial discount rate

# Cost annualisation and financial discount rate

The investment cost  $I_0$  needs to be recovered by payments in every year, which can be expressed as follows:

$$I_0 = \frac{I_A}{(1+r)^1} + \frac{I_A}{(1+r)^2} + \frac{I_A}{(1+r)^3} + \dots + \frac{I_A}{(1+r)^N}$$

$$I_0 = \frac{I_A}{(1+r)^1} + \frac{I_A}{(1+r)^2} + \frac{I_A}{(1+r)^3} + \dots + \frac{I_A}{(1+r)^N} \quad | \cdot (1+r) \text{ for geometric series}$$

$$(1+r)I_0 = I_A \left( \frac{1 - \frac{1}{(1+r)^N}}{1 - \frac{1}{1+r}} \right) \quad | \cdot \left( \frac{1}{1+r} \right) \quad | \cdot \left( 1 - \frac{1}{1+r} \right)$$

Then we obtain:

$$I_0 \left( 1 - \frac{1}{1+r} \right) = \frac{I_A}{1+r} - \frac{I_A}{(1+r)^{N+1}}$$

$r$  : financial discount rate

$N$  : number of operational years of the asset

$I_A$  : annualised cost

$I_0$  : capital cost, investment in the first year

# Cost annualisation and financial discount rate

$$I_0\left(1 - \frac{1}{1+r}\right) = \frac{I_A}{1+r} - \frac{I_A}{(1+r)^{N+1}} \quad | \cdot (1+r)^{N+1}$$

Then we get:

$$I_0((1+r)^{N+1} - (1+r)^N) = I_A((1+r)^N - 1)$$

Reordering the equation, we obtain:

$$I_A = I_0 \frac{(1+r)^N \cdot r}{(1+r)^N - 1}$$

**Cost annualisation**

$r$  : financial discount rate

$N$  : number of operational years of the asset

$I_A$  : annualised cost

$I_0$  : capital cost, investment in the first year

For investment in the energy sector, a financial discount rate of 7% is typically assumed.

# Including climate change in economic optimisation and social discount rate

# Including climate change in economic optimisation

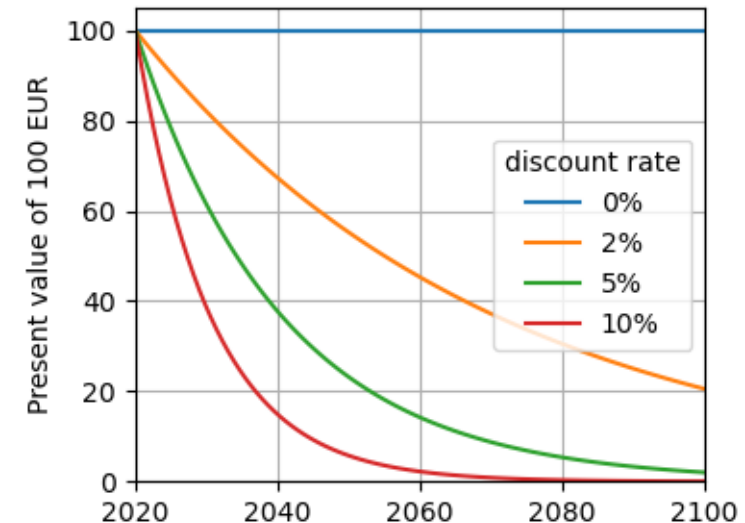
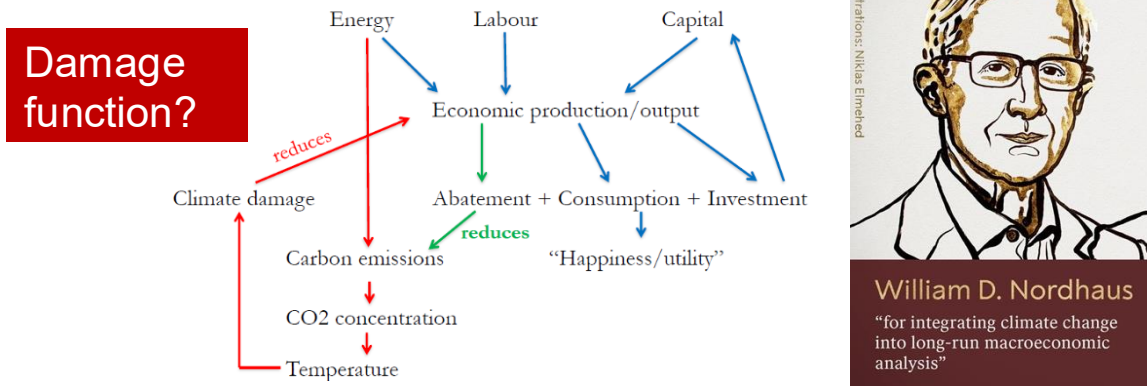
Initially, **cost-benefit analyses** were developed, aiming at evaluating to what extent the cost of climate change mitigation compensates for the avoided climate change impacts.

**Which temperature increase is cost-optimal (from an economic point of view)?**

The main challenges were:

1. How to quantify impacts of climate change through damage functions or using the social cost of carbon (SCC)
2. The assumed social discount rate determines the results

Figure 1: Schematic illustration of the DICE model



Stern/Nordhaus controversy:

- Nordhaus found optimal temperature increase 3.5°C ( $r = 4.5\%$ ), SSC of 10 USD/tCO<sub>2</sub> ( $r = 3\%$ )
- Stern: SSC of 85 USD/tCO<sub>2</sub> ( $r = 1.4\%$ )

D. Roberts, [Discount rates: A boring thing you should know about](#), 2012

# Including climate change in economic optimisation

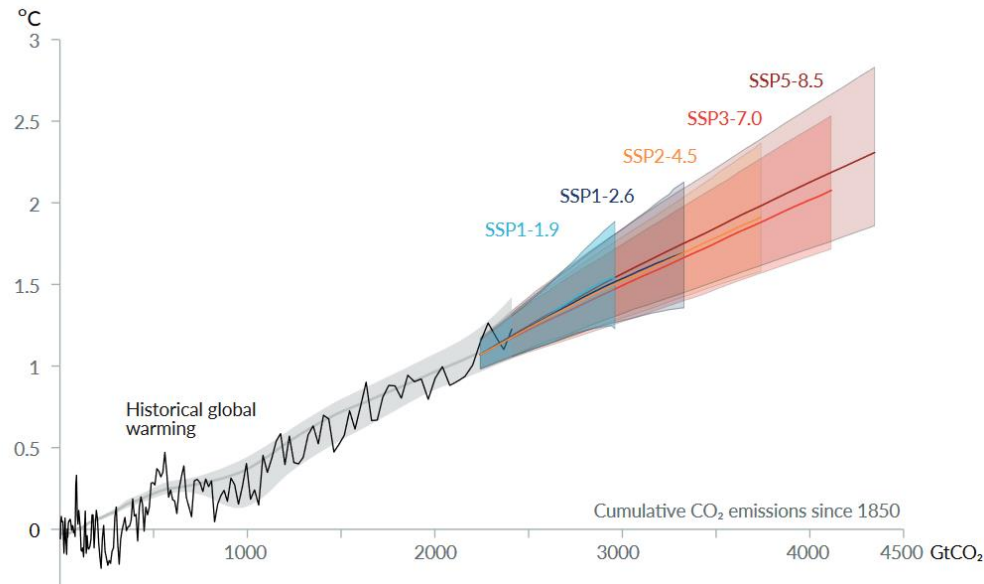
Later, **cost-effectiveness analyses** were developed, aim at estimating the most cost-effective strategy to attain a climate target.

**What is the best way of transforming the system while keeping emissions below CO<sub>2</sub> budget?**

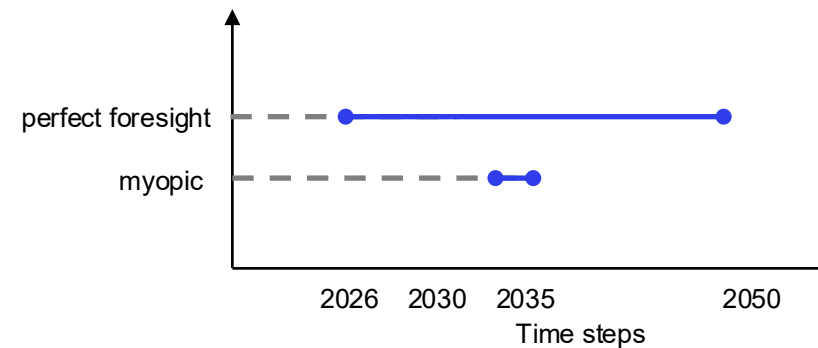
We can impose the CO<sub>2</sub> constraint throughout a transition path to limit cumulative emissions

$$\sum_{s,t} \text{CO}_2 \text{ emissions} \leq \text{CO}_2 \text{ limit} \leftrightarrow \mu_{\text{CO}_2}$$

$$\int_{2026}^{2050} \sum_{s,t} \text{CO}_2 \text{ emissions} \leq \text{CO}_2 \text{ budget}$$



We can assume perfect-foresight or myopic transition.



The perfect-foresight approach obtains the optimal transformation path and is influenced by the assumed social discount rate. The myopic approach captures the short-sighted behaviour of policy-makers and the sunk costs of bad decisions.

- 90% GHG emission target for 2040, with 5% via international credits
  - Permanence?
  - Leakage
  - Additionality (similar to permits of renewable electricity)
- Increased competition
- Industrial policy
- Carbon border tax?

## 2040 climate target: Council gives final green light

Today, the Council formally adopted the amended **European climate law**, introducing a binding intermediate climate target, for 2040, of a **90% reduction in net greenhouse gas (GHG) emissions** compared to 1990 levels. This new target strengthens the EU's path towards achieving **climate neutrality** by 2050 across all sectors of the economy.

From 2036 onwards, **high-quality international credits** may be used **up to a limit of 5%** of 1990 EU net emissions to make an adequate contribution towards the 2040 target in a way that is both ambitious and cost-efficient. This means that at least **85%** of emissions reductions must be achieved within the EU. Credits must be based on credible activities of GHG reduction in partner countries, in line with the Paris agreement.

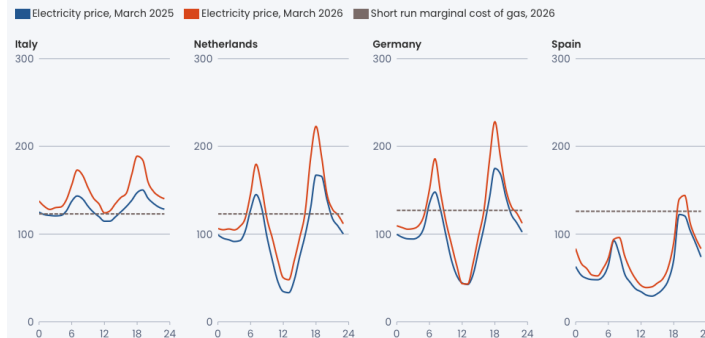
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## How Europe sleepwalked into yet another energy crisis

### Spain's electricity prices remain below gas power costs, while EU countries with more gas exposure see price spikes

Average hourly weekday prices, 1-10 March 2026 vs March 2025 (€/MWh)



EMBER

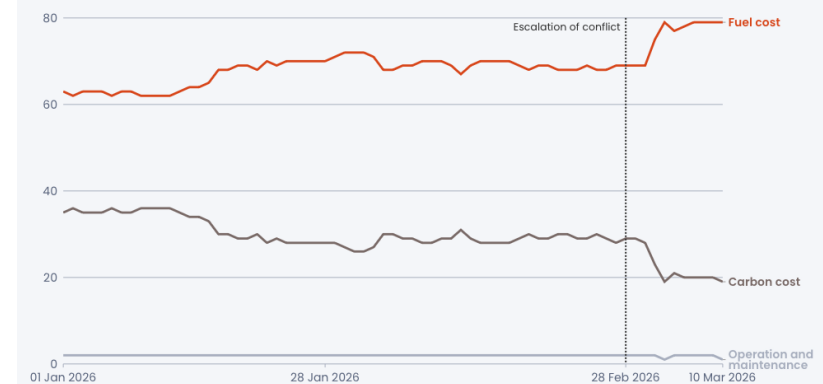
Carbon offsetting

This article is more than 5 months old

## Carbon offsets fail to cut global heating due to 'intractable' systemic problems, study says

### Carbon costs account for a relatively small share of gas-fired power costs in the EU

Share of short run marginal cost of gas-fired power generation in the EU (%)



EMBER

# Problems for this lecture

Problems 9.1 + 9.3 (**Group 19**)

Problems 9.2 (**Group 20**)

Additional recommendations:

[The Economist: How do carbon markets work](#)

[Externalities in 90 seconds](#)

[Think Econ: Externalities](#)

[IMF: Back to Basics // Externalities](#)

# DTU

